DIPLOMA PROGRAMME PRACTICE / HL / SET A

Economics / Paper 2

Worked mark scheme · 105 minutes · 40 marks

Choose one of the two cases and answer all parts (a) to (g) of that case. A calculator is permitted. Use the extracts and data in your answers. Show workings and units, and draw fully labelled diagrams on separate answer paper.

Judge extended responses as a whole using the markband for that part. Award valid alternative analysis and diagrams. In linked calculations, credit correct use of an earlier result. The suggested judgements are examples.

HOW TO MARK THIS PAPER

Use the question-specific marking notes for parts (a) to (f). Apply the 15-mark scale to part (g). Only the selected case contributes to the score.

15-MARK SCALE

0: no relevant economics. 1–3: assertions or isolated concepts. 4–6: some relevant explanation, limited application. 7–9: coherent analysis with some application and consideration of consequences. 10–12: developed, applied analysis weighing competing considerations with a supported view. 13–15: precise theory and relevant diagrams where useful, evidence integrated throughout, and a balanced judgement that explains conditions or relative importance. Other defensible conclusions can receive full credit.

QUESTIONS IN THIS SCHEME

1 Water security and rural incomes in Morocco [40]

2 Chile and a copper revenue shock [40]

QUESTION 1 / 40 MARKS

Water security and rural incomes in Morocco

Background reading. World Bank, Morocco water security and resilience programme (2023)

PART a(i) / 2 MARKS

Define a common pool resource in the context of the aquifer in Text A.

Award. Consumption is rival: extraction leaves less available to others (1). Exclusion of potential users is difficult or costly (1). An example alone does not establish both characteristics.

PART a(ii) / 2 MARKS

Define opportunity cost as it applies to the treasury's choice in Text B.

Award. The value or benefit of the next best alternative (1) sacrificed by making a choice (1). Here it may be the benefits forgone from the other water programme.

PART b(i) / 3 MARKS

Calculate the percentage change in metered withdrawals and the percentage change in the water charge, using initial values.

Working. Withdrawal change = (36 − 40)/40 × 100 = −10%. Charge change = (1.50 − 1.20)/1.20 × 100 = +25%. Award one mark for valid substitution and one for each signed result or clearly stated direction.

PART b(ii) / 2 MARKS

Calculate the revenue collected from the after-pilot charge, assuming all metered withdrawals are billed and paid.

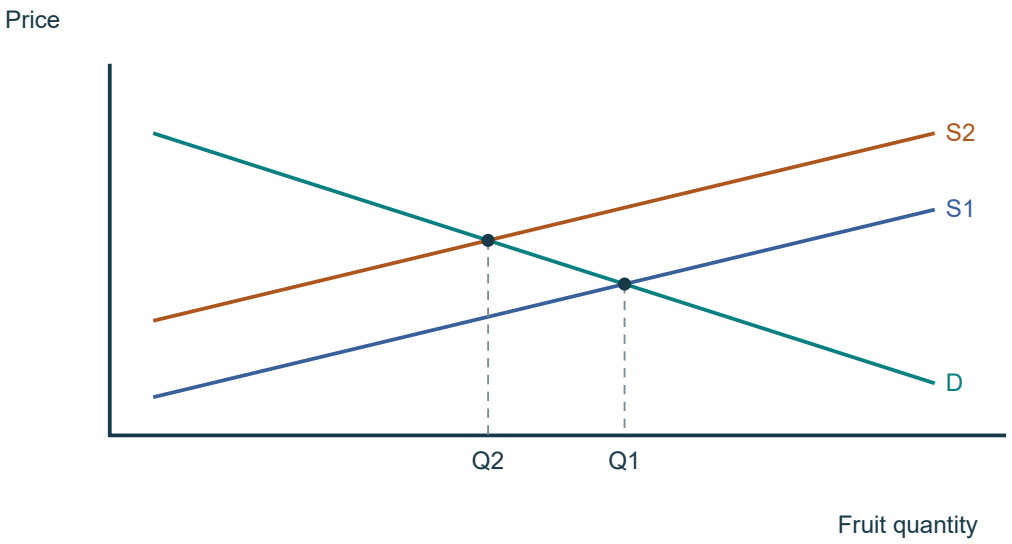
Working. 36 million m³ × 1.50 dirhams/m³ (1) = 54 million dirhams per season (1). Revenue is not net social benefit.

PART c / 4 MARKS

Using a demand and supply diagram, explain how a per-unit abstraction charge paid by growers could affect the supply of orchard fruit.

Diagram and explanation. Use fruit price and fruit quantity axes. Label initial demand and supply and the new supply curve to the left or above the original (up to 2). Water is an input; the charge raises marginal costs. At unchanged demand the fruit price rises and equilibrium output falls, other things equal (up to 2).

WORKED REPRESENTATION / c



PART d / 4 MARKS

Explain why credit constraints may cause a uniform equipment subsidy to benefit owners of large orchards more than tenant farmers.

Explanation. The subsidy lowers the private purchase cost, but the grower must finance the remainder. Large growers can borrow or use retained funds. A short lease limits a tenant's time to recover the investment. Relative uptake and total subsidy received can therefore be greater for established owners. Award up to four for a linked explanation grounded in Text B.

PART e / 4 MARKS

Using a production possibilities diagram, explain how a lasting deterioration in reliable water supply can change the attainable combination of farm and other output.

Diagram and explanation. Label farm output and other output; draw an initial frontier and a new frontier inward, particularly on the farm-output intercept if agriculture relies more heavily on water (up to 2). Lost productive capacity reduces feasible combinations; distinguish a new frontier from movement caused by temporary underuse (up to 2).

PART f / 4 MARKS

Explain why a reduction in water used per hectare may fail to reduce total withdrawals in the long run.

Explanation. Total use equals water per hectare times area. Cheaper effective irrigation can raise returns and encourage expansion. The trial area grew from 10,000 to 10,500 hectares although total metered use fell. Continued expansion or diversion to unmetered wells could offset technical savings. Four marks for mechanism, application and a conditional conclusion.

PART g / 15 MARKS

Using the texts, data and your knowledge of economics, evaluate whether Morocco's practice district should prioritise a metered abstraction charge or an equipment subsidy to protect groundwater and rural livelihoods.

Worked argument. Charging raises the marginal cost of an additional withdrawal. The pilot's 25% price rise was associated with 10% less metered water: some demand responded, but the smaller quantity response means the burden on users remains material. The 54 million dirhams collected could fund meters and connections, subject to installation and collection costs.

Counterargument. Subsidised drip equipment may preserve yields for each unit of water without an immediate income shock. One-season tenants might be unable to afford even the subsidised price. Cheaper irrigation can expand acreage, and charging only metered users can instead move pumping to unmetered wells. A before/after pilot with similar rainfall does not identify the nationwide response.

Supported judgement. Prioritise metering and a charge in areas where coverage can be enforced, with grants targeted to vulnerable households. A subsidy may be the better initial policy where monitoring is infeasible and assistance reaches tenants. Compare aquifer levels, total withdrawals and farm income over several seasons before scaling up.

Using Figure 1. The illustrative metering rates are 38% for small farms, 64% for medium farms and 89% for large farms. Unequal coverage means the pilot cannot stand in for all withdrawals.

Apply the 15-mark scale on the first page to this response as a whole.

QUESTION 2 / 40 MARKS

Chile and a copper revenue shock

Background reading. IMF, Fiscal Considerations in Managing Stabilization Funds in Chile (2023)

PART a(i) / 2 MARKS

Define an export.

Award. A good or service supplied by a resident of one economy (1) to a non-resident/overseas buyer (1). A transfer of money alone does not define an export.

PART a(ii) / 2 MARKS

Define a budget deficit.

Award. Government expenditure exceeds government revenue (1) during a given period (1). Distinguish the annual flow from outstanding public debt.

PART b(i) / 3 MARKS

Calculate copper export revenue in each scenario and the change, in billions of US dollars.

Working. Baseline: 8,400 × 4.0 million = US$33.60 billion (1). Shock: 7,000 × 4.0 million = US$28.00 billion (1). Revenue falls by US$5.60 billion (1).

PART b(ii) / 2 MARKS

Calculate the shock-year budget deficit per resident.

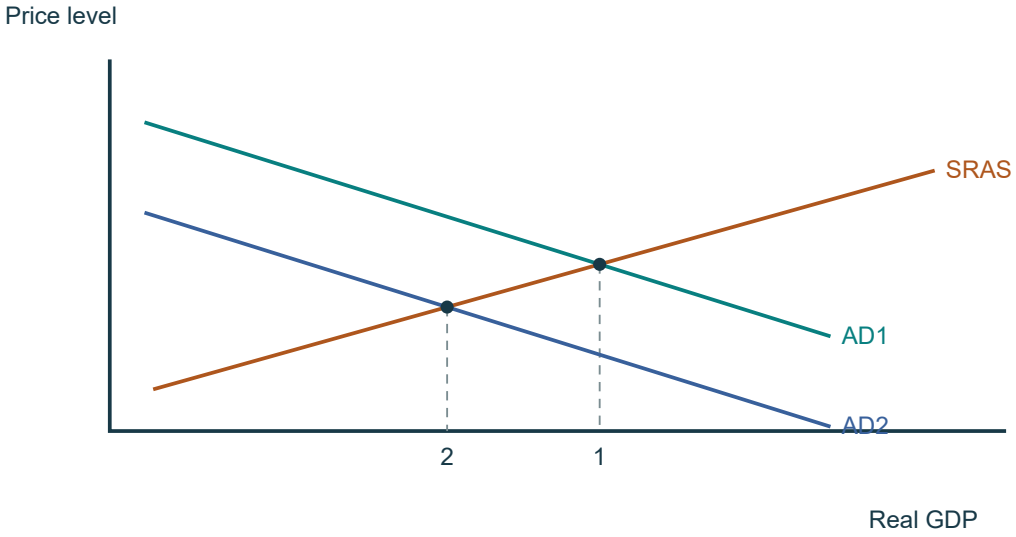
Working. (82 − 75) billion / 20 million (1) = US$350 per resident for the year (1). This average is not a tax bill for each individual.

PART c / 4 MARKS

Using an AD–AS diagram, explain how the loss of export receipts could reduce real output in the short run.

Diagram and explanation. Label price level and real GDP, SRAS and AD; shift AD left and indicate lower equilibrium output (up to 2). Lower net exports cut expenditure; weaker incomes and local orders can depress spending further. Assume spare capacity and other spending unchanged (up to 2).

WORKED REPRESENTATION / c



PART d / 4 MARKS

Explain why the fall in government revenue need not equal the fall in copper export revenue.

Explanation. Export revenue accrues initially to producers. Royalties and taxes capture only portions, sometimes on profit rather than total sales. Other taxes or non-mining activity can change. Here public revenue falls US$5 billion while copper receipts fall US$5.6 billion; neither number gives the royalty rate. Reward a complete mechanism with evidence.

PART e / 4 MARKS

Using a labour market diagram for local repair workers, explain a possible effect of postponing school repairs.

Diagram and explanation. Label wage and labour employed, labour supply and derived demand. Shift labour demand left with lower equilibrium employment (up to 2). Fewer contracts reduce the extra revenue from employing workers; wages may fall, or sticky wages may instead increase unemployment (up to 2).

PART f / 4 MARKS

Explain why improving occupational mobility may be insufficient to secure employment for the mining town's displaced workers.

Explanation. Training changes the tasks workers can perform. Jobs may still be in distant towns with high moving or housing costs. Training takes two years and income is needed now. If demand is weak across industries, qualified workers can remain unemployed. Reward a linked explanation of at least one distinct remaining constraint.

PART g / 15 MARKS

Using the texts, data and your knowledge of economics, evaluate the use of a stabilisation fund to maintain public spending after the fall in the copper price.

Worked argument. With fixed volume, US$5.6 billion of copper receipts disappears. Saving in a boom and drawing down during a temporary fall can prevent simultaneous cuts to public contracts and private incomes. Maintaining repairs protects jobs and school services. The projected deficit is US$7 billion, not the value of the export loss. Fund assets finance a deficit; they do not restore ongoing tax revenue.

Counterargument. Imported equipment means some support raises overseas output. A drawdown has an opportunity cost and can deplete assets if the price fall persists. Continuing every low-value project may displace a training programme that reduces longer-term dependence. Training alone offers little immediate relief where vacancies are distant.

Supported judgement. Protect high-value maintenance temporarily if the fund is liquid and the shock likely cyclical. Publish a drawdown rule and reassess the sustainable budget if copper income remains low. Evidence on the fund balance, future prices and local employment would change the recommended scale.

Using Figure 1. Illustrative domestic purchasing shares are 72% for repairs, 28% for new haul trucks and 86% for training. A larger domestic share limits initial import leakage but does not establish which project creates most benefit.

Apply the 15-mark scale on the first page to this response as a whole.